

9. (a) Addition of a solution containing $\text{Ag}^+(\text{aq})$ to solutions containing halide ions such as chloride, bromide or iodide causes precipitates to form.

- (i) Give the colours of the precipitates formed with chloride, bromide and iodide ions. [1]

chloride ions

bromide ions

iodide ions

- (ii) The standard enthalpy change of formation of $\text{AgCl}(\text{s})$ is $-127.1 \text{ kJ mol}^{-1}$.

Use the data below to calculate the standard enthalpy change of formation of $\text{AgBr}(\text{s})$ and hence show which of the two silver halides is more stable with respect to its elements. You **must** show your working. [4]

Reaction	Standard enthalpy change, $\Delta H^\theta / \text{kJ mol}^{-1}$
$\text{Ag}(\text{s}) \longrightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$	+105.6
$\text{Br}_2(\text{l}) + 2\text{e}^- \longrightarrow 2\text{Br}^-(\text{aq})$	-243.1
$\text{AgBr}(\text{s}) \longrightarrow \text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq})$	+84.4

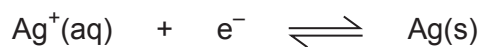
.....

.....

.....



- (b) The standard electrode potential for the half-equation below is +0.80 V.



- (i) When a piece of copper metal is placed in a solution of silver nitrate, a displacement reaction occurs.

I. Write the **ionic** equation for this displacement reaction. [1]

.....

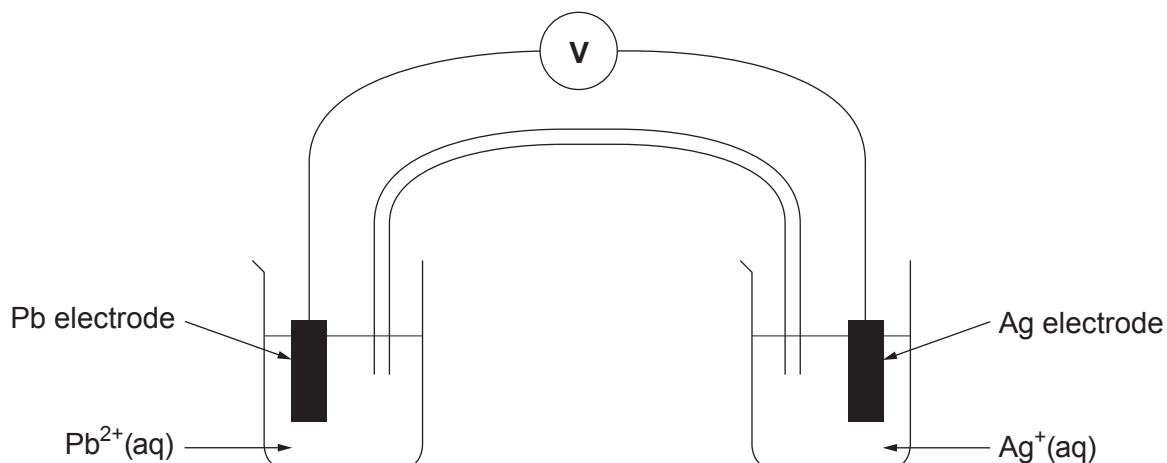
II. State what information this provides regarding the standard electrode potential for the $\text{Cu}(\text{s})|\text{Cu}^{2+}(\text{aq})$ half-cell. Give a reason for your answer. [2]

.....

.....

.....

- (ii) The apparatus below was assembled with the $\text{Pb}(\text{s})|\text{Pb}^{2+}(\text{aq})$ half-cell under standard conditions connected to the $\text{Ag}(\text{s})|\text{Ag}^+(\text{aq})$ half-cell. The silver is the positive electrode.



The value on the high-resistance voltmeter was recorded with different concentrations of silver ions used in the $\text{Ag(s)}|\text{Ag}^+(\text{aq})$ half-cell. The results are shown below.

Concentration of $\text{Ag}^+(\text{aq})$ / mol dm^{-3}	Value recorded on high-resistance voltmeter / V
1.0	0.93
0.1	0.87
0.01	0.81
0.001	0.75
0.0001	0.69

- I. Calculate the value of the standard electrode potential for the $\text{Pb(s)}|\text{Pb}^{2+}(\text{aq})$ half-cell.

[2]

$$E^\theta = \dots\dots\dots \text{ V}$$

- II. Explain why the value recorded on the voltmeter becomes less positive as the concentration of silver ions decreases.

[3]

.....

.....

.....

.....

